

The first theorem states that the complement of a product is equal to the sum of the complement i.e. $\overline{A \cdot B} = \overline{A} + \overline{B}$

The second theorem states that the complement of a sum is equal to the product of the complements i.e. $\overline{A + B} = \overline{A} \cdot \overline{B}$

Proof for De Morgan's theorem by Perfect Induction method.

	1	2	3	4	5	6	7	8	9	10
A	B	$\overline{A}$	$\overline{B}$	$A + \overline{A} \cdot B$	$A + \overline{A} \cdot \overline{B}$	$\overline{A + B}$	$\overline{A} \cdot \overline{B}$	$\overline{A + B}$	$\overline{A} \cdot \overline{B}$	$\overline{A + B}$
0	0	1	1	0	0	1	1	1	1	1
0	1	1	0	1	0	0	0	1	1	1
1	0	0	1	1	0	0	0	1	1	1
1	1	0	0	1	1	0	0	0	0	0

from the above equal. Therefore

$$\frac{A+B}{A+B} = \overline{A \cdot B} \quad \text{Columns 7 and 8 are}$$

Similarly, column 9 and 10 are equal
Therefore $\overline{A \cdot B} = \overline{A} + \overline{B}$.

Minimization (Simplification) of Boolean
Expressions using algebraic method

The switching of Boolean expressions can be simplified by applying properties, laws and theorems of Boolean Algebra. See examples below:

Prove that $AB + BC + \overline{B}C = AB + C$
Solution

$$AB + BC + \overline{B}C = AB + C(B + \overline{B})$$

$$\because B + \overline{B} = 1$$

$$= AB + C(1)$$

$$= AB + C$$

? Simplify the expression

Solution

$$Y = (\bar{A} + B)(A + B)$$

$$Y = (\bar{A} + B)(\bar{A} + B)$$

$$= \bar{A} \cdot \bar{A} + \bar{A} \cdot B + A \cdot B + B \cdot B$$

$$\text{Idempotence: } \bar{A} \cdot \bar{A} = 0, \bar{A} + A = 1, B \cdot B = B$$

$$= \bar{A} \cdot \bar{A} + \bar{A} \cdot B + A \cdot B + B \cdot B$$

$$= 0 + B(\bar{A} + A) + B$$

$$= B(1) + B$$

$$\text{Idempotence} = B + B$$

$$=$$

$$B$$

Simplify

$$Y = \bar{A} \bar{B} \bar{C} + \bar{A} B \bar{C} + A \bar{B} \bar{C} + A B \bar{C}$$

Solution

$$Y = \bar{A} \bar{C}(\bar{B} + B) + A \bar{C}(\bar{B} + B)$$

$$= \bar{A} \bar{C}(1) + A \bar{C}(1)$$

$$= \bar{A} \bar{C} + A \bar{C}$$

$$= \bar{C}(\bar{A} + A)$$

$$= \bar{C}(1)$$

$$= \bar{C}$$

$$= \bar{C}$$

4. Complement the expression $\bar{A}B + C\bar{D}$
solution

$$\begin{aligned}\overline{\bar{A}B + C\bar{D}} &= (\overline{\bar{A}B}) \cdot (\overline{C\bar{D}}) \\ &= (\bar{\bar{A}} + \bar{\bar{B}}) \cdot (\bar{C} + \bar{\bar{D}}) \\ &= (A + B) \cdot (\bar{C} + D) \text{ per}\end{aligned}$$

MINTERM AND MAXTERM

Boolean expressions

1. The sum-of-products come in two forms:

MINTERM Form, and it is also called Product term in Boolean function, in which

every element present is either in normal or in complemented form e.g. $\bar{A} \cdot \bar{B} + A \cdot B = Y$

$\bar{A} \cdot \bar{C} + A \cdot B \cdot C = Y$, $A \cdot 0 + \bar{B} \cdot \bar{D} + C \cdot \bar{D} = Y$ etc.

2. The Product-of-sums form which is also called MAXTERM FORM, and it is defined

every element present is either in normal or in complemented form e.g. $(A+B) \cdot (\bar{A}+\bar{B}) = Y$

$(\bar{A}+B) \cdot \bar{C} = Y$, $(A+B) \cdot (\bar{C}+\bar{D}) \cdot (A+C) = Y$ etc.

Circuits

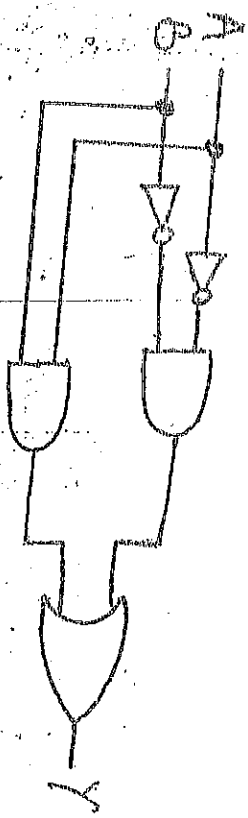
can be constructed from MINTERM and MAXTERM Boolean expressions, using

the examples in 1 and 2 above.

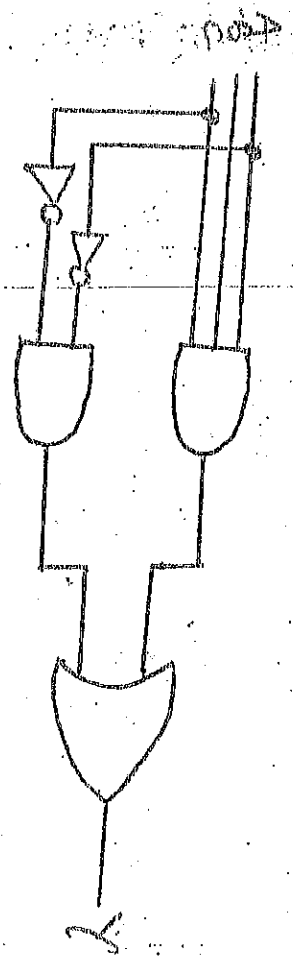
Drawing circuits from a MINTERM Boolean

expression is sum-of-products form.

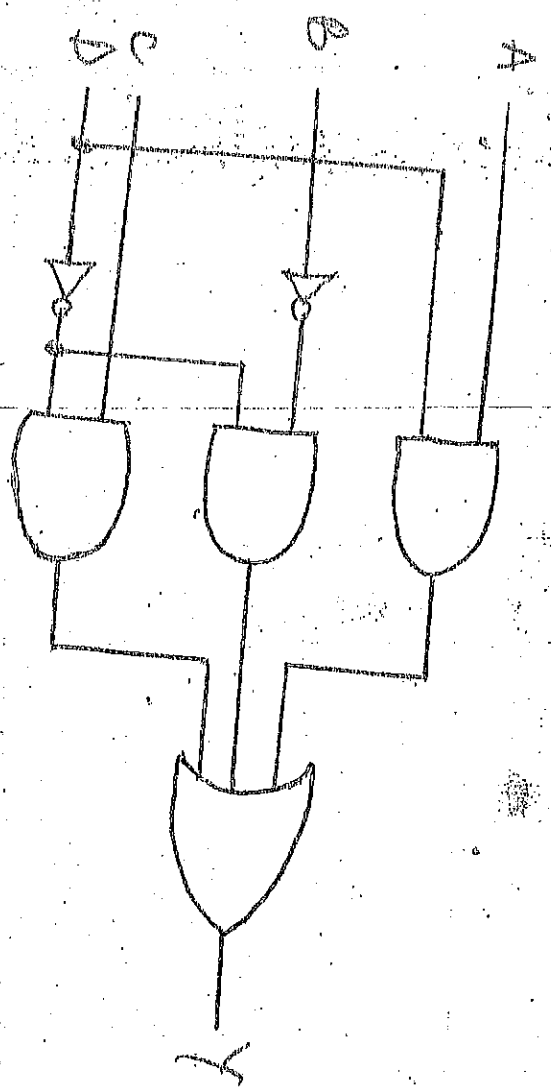
$$A \cdot B + A \cdot B = Y$$



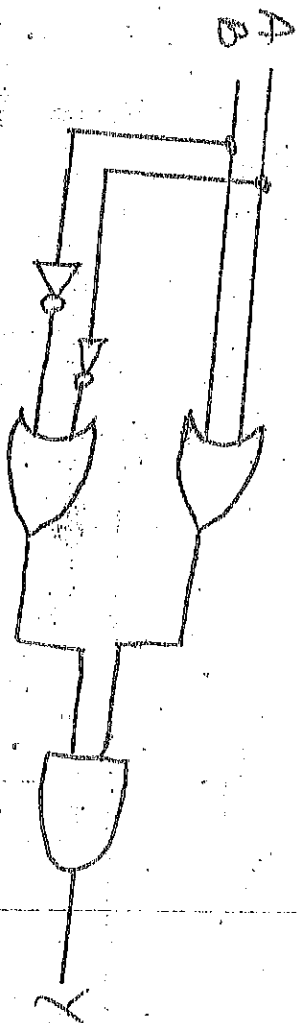
$$\bar{A} \cdot \bar{C} + A \cdot B \cdot C = Y$$



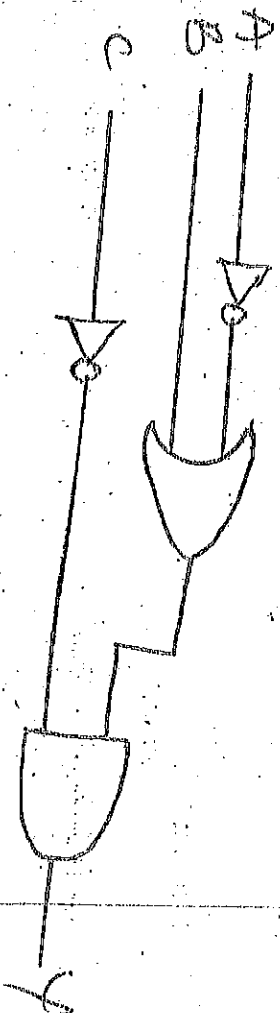
$$A \cdot D + \bar{B} \cdot \bar{D} + C \cdot \bar{A} = Y$$



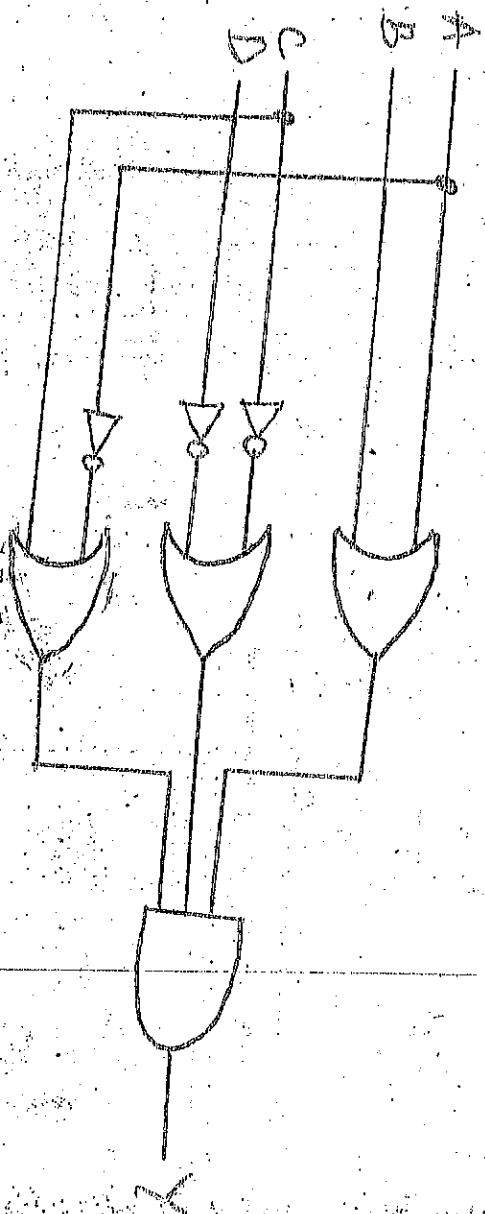
∴ we can derive sum of products from a MAXTERM Boolean expression i.e. products-of-sums form.



6) $(\bar{A} + B) \cdot \bar{C} = Y$



7) $(A + B) \cdot (C + \bar{D}) \cdot (C\bar{A} + C) = Y$



Boolean Expressions AND Truth Tables

Boolean expressions are convenient method of ~~not~~ describing how a logic circuit operates. The truth table is another precise method of describing how logic circuit works.

Truth tables to Boolean expression

Using the table below, write the min term

Boolean expression.

Input (switches)			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

$$\overline{A} \cdot B \cdot \overline{C}$$

$$A \cdot \overline{B} \cdot C$$

$$\overline{A} \cdot B \cdot \overline{C} + A \cdot \overline{B} \cdot C = Y$$

Using the truth table below, write the minterm Boolean expression

Input (switches)			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$A \cdot \bar{B} \cdot C$

$A \cdot B \cdot C$

Ans: $A \cdot \bar{B} \cdot C + A \cdot B \cdot C = Y$

$A [\bar{B} \cdot C + B \cdot C]$

DA

Support ~~Exposition~~ of TRUTH TABLE

Can other output = 0. Write the

sum - of - product Boolean expression for this situation using three inputs (switches)

Input (Switches)			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

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$$\frac{AB}{\sqrt{u^2 + \frac{1}{u^2}}}$$

5

assuming that only line 4 and 5 of the truth table produces an output of 1 (all other outputs = 0). Write the sum-of-products boolean expression for this situation.

Input			Output
A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

$\overline{A} \cdot B \cdot C$
 $A \cdot \overline{B} \cdot \overline{C}$

$$\overline{A} \cdot B \cdot C + A \cdot \overline{B} \cdot \overline{C} = Y$$

KARNAUGH MAP

Karnaugh map (K-map) is a pictorial method used to minimize Boolean expressions without having to use Boolean algebra theorems and laws for manipulations.

Examples:

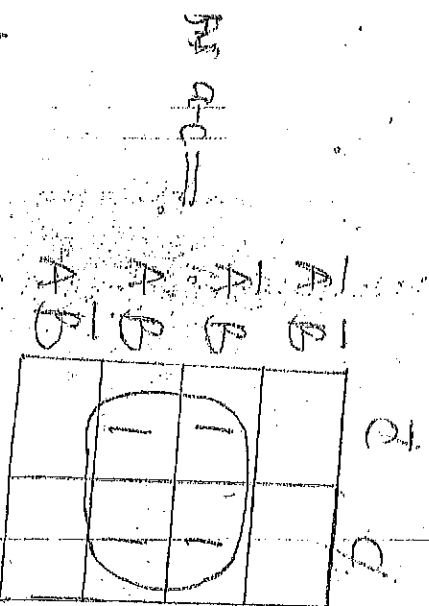
Simplifying the Boolean expression

$$\bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot B \cdot \bar{C} + A \cdot B \cdot C = Y \text{ by } -$$

Plotting is on a three-variable Karnaugh map

Looping groups of two or four. Is to eliminate variables whose complement appears within the loops.
Writing the simplified minterm Boolean expression

Solution:



d. $B = 1$

2. Simplify the Boolean expression

$$\bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + A\bar{B}C = Y \text{ by}$$

a. Plotting 1s on three variable Karnaugh map

b. Looping groups of two or four 1s

c. Eliminating variables whose complement appears within the loops

d. Writing the simplified Boolean expression.

Solution:

Ans a-c =

	$\bar{C}$		C
$\bar{A}\bar{B}$			1
$\bar{A}B$			1
AB			
$A\bar{B}$	1	1	

d. $\bar{A}\bar{B}C + A\bar{B} = Y$

1. Simplify the Boolean expression

$$\bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D} + \bar{A}B\bar{C}D + A\bar{B}C\bar{D} + A\bar{B}C\bar{D} = Y \text{ by}$$

a. Plotting 1s on a four-variable Karnaugh map

b. Grouping groups of two or four 1s
c. Eliminating variables whose complements appear within loops

d. Merging the simplified minterm Boolean expression

Solution:

$$\text{Ans } A = C = \bar{C}$$

Ans C

	$\bar{C}\bar{D}$	$\bar{C}D$	$C\bar{D}$	CD
$\bar{A}\bar{B}$				
$\bar{A}B$	1	1		
$A\bar{B}$	1	1		
AB			1	1

d. $B \cdot \bar{C} + A \cdot B \cdot C = Y$

BINARY BITS, BYTES, ALIBITS,
WORDS AND BINARY CODED DECIMAL.

Word :- A word is a combination of 16 bits, 32 bits or 64 bits depending on the computer.

Kilobyte (kb) :- A kilobyte has a memory unit of 1024.

Megabyte :- A megabyte has a unit of 1,048,576. This is approximately 1 million bytes.

Gigabyte (gb) :- A gigabyte has a memory unit of 1,073,741,824. This is approximately 1 billion bytes.

Terabyte :- A terabyte has a memory unit of approximately 1 trillion bytes.

Bit - A bit (short for binary digit)

is the smallest unit of information that can be stored in a computer.

A bit has a single binary value, either 0 or 1. Bits in computers are grouped to form a larger unit of information.

Byte - A byte is a combination of eight bits. Eight bits represent a character and is called a byte.

Nibble - A nibble is a combination of four bits, in other words a nibble is half a byte.

BINARY-CODED DECIMAL (BCD) DECODE

Definition:

The binary-coded decimal (BCD) is an encoding for decimal numbers in which each digit is represented by its own binary sequence. Its main virtue is that it allows easy conversion to decimal digits for printing or display and faster decimal calculations. Its drawbacks are the increased complexity of circuits needed to implement mathematical operations and a relatively inefficient encoding.